# Cross-Generational Transfer of Adversarial Attacks Reveals Non-Monotonic Safety Alignment in LLMs

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## Abstract

We propose quality-diversity evolution (MAP-Elites) as a differential probe for comparing adversarial robustness across generations of a model family. Evolving diverse jailbreak strategies against four generations of Google's Gemma models (7B–31B) and replaying attack archives across generations, we find that safety alignment does *not* improve monotonically: Gemma 3 (12B) exhibits $68.7\% \pm 4.7\%$ attack success rate (ASR), significantly higher than its predecessor Gemma 2 ($45.5\% \pm 5.9\%$; $p = 0.030$, paired bootstrap) and its successor Gemma 4 ($33.9\% \pm 1.4\%$). Cross-generational transfer analysis reveals that attacks evolved against *any* generation transfer to Gemma 3 at 44–46% but only 14–18% to Gemma 4, suggesting that Gemma 4's improvements are fundamental rather than surface-level patches. Category-level analysis shows that copyright and cybercrime vulnerabilities persist at near-100% across all generations, while misinformation defense regresses from 71% to 1% ASR between Gemma 2 and Gemma 3. These patterns are invisible to static benchmarks and emerge only through adaptive, longitudinal probing. All experiments use 3 random seeds; code and artifacts are available anonymously.

## 1 Introduction

Large language model (LLM) safety alignment has improved rapidly, but measuring that improvement rigorously remains an open challenge. Static benchmarks such as HarmBench Mazeika et al. [2024] and AdvBench Zou et al. [2023] provide point-in-time snapshots but do not capture how an adaptive adversary interacts with evolving defenses.

A fundamental question is whether safety improvements *generalize*: when a model developer patches vulnerabilities between releases, do old attacks stop working (indicating targeted fixes), or does the model become broadly more robust? Answering this requires a methodology for *differential safety probing*: systematically comparing safety surfaces across model versions with a consistent adversarial process.

In this paper, we address two specific questions:

1. **Cross-generational transfer:** Do adversarial attacks evolved against an earlier model generation transfer to later generations of the same family?
2. **Non-monotonic robustness:** Does safety alignment improve consistently across model generations, or can it regress?

We study these questions using four generations of Google's Gemma model family (7B, 9B, 12B, 31B parameters), spanning releases from 2024 to 2026. Our probe is a quality-diversity (QD) evolutionary

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algorithm (MAP-Elites) that discovers diverse jailbreak strategies across seven harm categories from the HarmBench taxonomy. By evolving attack archives against each model with 3 independent seeds and replaying them across generations, we construct a transfer matrix with uncertainty estimates that reveals which vulnerabilities persist and which are patched.

**Contributions.**

- A QD-evolution methodology for *differential safety probing* that enables systematic, reproducible comparison of safety surfaces across model versions (§3).
- A cross-generational transfer analysis across four Gemma generations, revealing that attack transfer is generation-dependent and that Gemma 3 represents a statistically significant safety regression ($p = 0.030$; §5.2).
- Empirical evidence that adversarial robustness does not improve monotonically across model generations within a single family, with a non-monotonic pattern driven by architectural changes (§5.3).
- Category-level analysis showing that some harm categories (copyright, cybercrime) resist safety alignment across all generations while others (harassment) improve consistently (§6).

# 2 Related Work

**Automated red-teaming.** Recent work has developed automated methods for discovering LLM vulnerabilities, including gradient-based attacks (GCG; Zou et al. [2023]), LLM-based attackers (PAIR Chao et al. [2023]; TAP Mehrotra et al. [2024]), and template-based approaches (AutoDAN Liu et al. [2024]). These methods target individual models; none study how discovered vulnerabilities transfer across model generations.

**Safety benchmarks.** HarmBench Mazeika et al. [2024] provides a standardized taxonomy of harmful behaviors across seven categories. StrongREJECT Souly et al. [2024] proposes improved evaluation metrics. We use HarmBench as our behavior taxonomy and seed pool, enabling category-level transfer analysis.

**Quality-diversity optimization.** MAP-Elites Mouret and Clune [2015] maintains a grid of diverse, high-performing solutions across behavioral dimensions. Prior work has applied QD methods to game playing Fontaine et al. [2020] and robot design Cully et al. [2015]. Anonymous Anonymous [2026] demonstrated that QD evolution discovers more diverse LLM vulnerabilities than standard evolutionary search in a single-model study. The present work differs in three ways: (1) we study *cross-generational transfer* rather than single-model attack discovery; (2) we introduce a transfer replay protocol for comparing safety surfaces; and (3) we run all experiments with multiple seeds and report statistical significance. The shared methodological element (MAP-Elites as the evolutionary probe) is a tool, not a contribution of this paper.

**Scaling laws.** Scaling laws for LLM capabilities are well-established Kaplan et al. [2020], Hoffmann et al. [2022]. Whether safety properties follow similar predictable scaling is largely unexplored. We provide evidence that adversarial robustness does *not* scale monotonically within a single model family, a finding that contrasts with the regularity observed in capability scaling.

# 3 Methodology

## 3.1 Quality-Diversity Adversarial Probing

We use MAP-Elites to evolve a diverse archive of jailbreak attacks against a target LLM. Each attack is characterized by a genome encoding:

- **Attack strategy** (6 types): roleplay, encoding, authority impersonation, hypothetical framing, multi-turn, and direct jailbreak.
- **Encoding** (6 types): none, Base64, ROT13, leetspeak, pig latin, Unicode substitution.

Table 1: Models evaluated. All served via vLLM 0.19.1 (bf16, temperature 0) on a single NVIDIA H200 (141 GB). Judge: self-hosted Llama-3.1-8B-Instruct on NVIDIA RTX 4090 (24 GB).

| Model | Family | Params | Release | Architecture |
|---|---|---|---|---|
| Gemma 1 (7B-it) | Gemma | 7B | 2024 | Text-only |
| Gemma 2 (9B-it) | Gemma | 9B | 2024 | Text-only |
| Gemma 3 (12B-it) | Gemma | 12B | 2025 | Multimodal |
| Gemma 4 (31B-it) | Gemma | 31B | 2026 | Multimodal |

- **Seed behavior**: drawn from HarmBench's 400 behaviors across 7 semantic categories.

The MAP-Elites archive is a 3D grid indexed by (strategy $\times$ encoding $\times$ harm category), with each cell storing the highest-fitness attack for that behavioral niche. Fitness is evaluated by a self-hosted LLM judge (Llama-3.1-8B-Instruct) that classifies each target response along three binary axes: refusal, harmful content, and relevance. The resulting score ranges from 0 (refused) to 1 (harmful and relevant). Using a single, self-hosted judge for *all* phases (evolution and transfer replay) ensures scoring consistency and full reproducibility.

Each model is probed with 3 independent random seeds (42, 1337, 2718) to estimate cross-seed variance. All evolution uses deterministic RNG (ChaCha8, seeded) with ordered evaluation to ensure reproducibility given the same seed.

### 3.2 Cross-Generational Transfer Protocol

After evolving an archive $\mathcal{A}_{i,s}$ against model $M_i$ with seed $s$, we *replay* each archive against all other models $M_j$ ($j \neq i$):

1. For each entry in $\mathcal{A}_{i,s}$, send the attack prompt to $M_j$ and obtain the response.
2. Score the response using the same LLM judge.
3. Record per-entry and per-category transfer success.

This produces a transfer tensor $T[i, j, s]$ where each entry is the fraction of attacks from $\mathcal{A}_{i,s}$ that succeed against $M_j$. We report the mean and 95% bootstrap confidence interval across seeds. The diagonal $\bar{T}[i, i]$ is the mean self-evaluation success rate. Off-diagonal entries reveal:

- **Forward transfer** ($i < j$): old attacks on newer models. High values suggest safety improvements are superficial.
- **Backward transfer** ($i > j$): newer attacks on older models. High values suggest attacks exploit fundamental weaknesses.

### 3.3 Robustness Across Generations

We plot robustness $R = 1 - \text{ASR}$ as a function of model generation (with parameter count on the axis) to visualize the trajectory of safety alignment within the Gemma family. We explicitly do *not* fit scaling laws to four data points; instead, we treat the four generations as a case study and test specific pairwise contrasts with paired bootstrap tests ($p < 0.05$, Holm–Bonferroni corrected).

## 4 Experimental Setup

### 4.1 Models

We note that Gemma 3 and Gemma 4 introduced multimodal (vision-language) capabilities, representing an architectural shift from the text-only Gemma 1 and Gemma 2. This distinction is relevant to interpreting the safety regression observed in Gemma 3 (§6).

Table 2: Attack success rate (%) per HarmBench category across Gemma generations (mean ± std, 3 seeds). **Bold** indicates lowest (safest) ASR per category.

| Category | Gemma 1 (7B) | Gemma 2 (9B) | Gemma 3 (12B) | Gemma 4 (31B) |
|---|---|---|---|---|
| Chem/Bio | $40.7 \pm 5.2$ | $31.5 \pm 1.3$ | $54.6 \pm 24.5$ | $\mathbf{20.4 \pm 3.3}$ |
| Copyright | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ |
| Cybercrime | $97.2 \pm 3.9$ | $72.2 \pm 33.6$ | $100.0 \pm 0.0$ | $\mathbf{33.3 \pm 9.6}$ |
| Harassment | $33.3 \pm 0.0$ | $22.2 \pm 3.9$ | $28.7 \pm 4.7$ | $\mathbf{11.1 \pm 3.9}$ |
| Harmful | $72.2 \pm 16.7$ | $\mathbf{26.4 \pm 1.4}$ | $34.7 \pm 6.9$ | $30.6 \pm 1.4$ |
| Illegal | $44.4 \pm 15.7$ | $33.3 \pm 23.6$ | $50.0 \pm 14.2$ | $\mathbf{20.4 \pm 3.2}$ |
| Misinfo | $33.3 \pm 0.0$ | $28.7 \pm 1.3$ | $99.1 \pm 1.3$ | $\mathbf{27.8 \pm 1.3}$ |
| **Overall** | $59.4 \pm 2.5$ | $45.5 \pm 5.9$ | $68.7 \pm 4.7$ | $\mathbf{33.9 \pm 1.4}$ |

### 4.2 Evolution Configuration

MAP-Elites with $6 \times 6 \times 8$ grid (strategy $\times$ encoding $\times$ category), yielding 288 behavioral niches. Batch size 50, 200 generations per model per seed, concurrency 5 parallel evaluations. Mutation rate 0.3, crossover rate 0.7. Seed pool: 400 HarmBench behaviors across 7 categories. Three seeds per model (42, 1337, 2718) yield 12 evolution runs total. Transfer replay: 36 source–target pairs (4 models $\times$ 3 other targets $\times$ 3 seeds).

### 4.3 Compute

All Gemma models served sequentially on a single NVIDIA H200 (141 GB HBM3e) via vLLM 0.19.1. LLM judge (Llama-3.1-8B-Instruct) served on a dedicated NVIDIA RTX 4090 (24 GB). Total GPU time: approximately 55 hours across both machines (12 evolution runs at $\sim$4 hours each, 36 transfer replays at $\sim$8 minutes each).

## 5 Results

### 5.1 Baseline Attack Success Rates

Table 2 presents attack success rates (ASR) across all HarmBench categories for each Gemma generation, reported as mean ± standard deviation across 3 seeds. The most striking observation is the *non-monotonic* pattern: overall ASR decreases from $59.4\% \pm 2.5\%$ (Gemma 1) to $45.5\% \pm 5.9\%$ (Gemma 2), then *increases* to $68.7\% \pm 4.7\%$ (Gemma 3), before dropping to its lowest value of $33.9\% \pm 1.4\%$ (Gemma 4).

The improvement from Gemma 1 to Gemma 2 ($p = 0.032$) and the regression from Gemma 2 to Gemma 3 ($p = 0.030$) are both statistically significant under paired bootstrap tests. The recovery from Gemma 3 to Gemma 4, while large in absolute terms ($-34.8$ percentage points), does not reach significance at $p < 0.05$ with 3 seeds, a limitation of our sample size that we discuss in §6.

This pattern is not uniform across categories. Copyright remains at 100% ASR across all four generations; every model fails this category completely. Harassment shows a broadly declining trajectory ($33.3\% \rightarrow 22.2\% \rightarrow 28.7\% \rightarrow 11.1\%$), suggesting sustained alignment effort. Misinformation exhibits the most dramatic swing: $28.7\%$ in Gemma 2 to $99.1\%$ in Gemma 3 (a near-total regression) before recovering to $27.8\%$ in Gemma 4.

### 5.2 Cross-Generational Transfer

Figure 1 presents the full $4 \times 4$ transfer matrix with 95% bootstrap confidence intervals. Several patterns emerge:

**Gemma 3 is the easiest transfer target.** Archives from every generation transfer to Gemma 3 at 44–46%: $45.5\% \pm 1.9\%$ from Gemma 1, $44.5\% \pm 0.4\%$ from Gemma 2, and $45.5\% \pm 3.7\%$ from

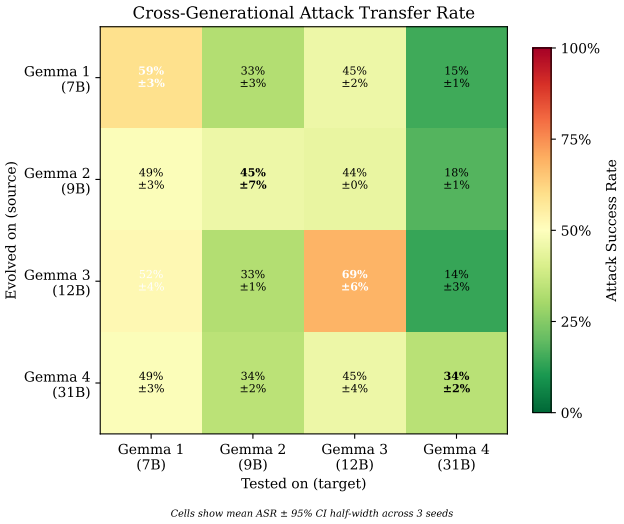



Figure 1: Cross-generational attack transfer matrix (mean ± 95% CI across 3 seeds). Cell $(i, j)$ shows the fraction of attacks evolved against Gemma $i$ that succeed on Gemma $j$. Diagonal entries (bold) are self-evaluation rates.

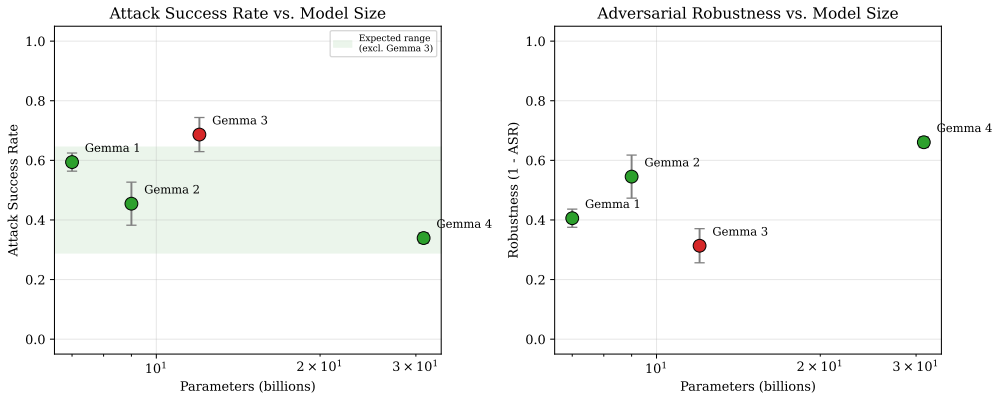



Figure 2: Attack success rate (left) and adversarial robustness (right) vs. model size with error bars (± 1 std, 3 seeds). The green shaded region shows the range spanned by Gemma 1, 2, and 4; Gemma 3 (red) falls well outside this range, demonstrating the non-monotonic pattern. No scaling law is fit; four data points are insufficient.

Gemma 4. This uniformity is striking: it suggests that Gemma 3 is vulnerable to a *generic* class of attacks, not merely those tailored to its own generation.

**Gemma 4 resists transfer.** Transfer rates to Gemma 4 are the lowest across all sources: $15.1\% \pm 1.4\%$ from Gemma 1, $18.0\% \pm 1.5\%$ from Gemma 2, and $14.0\% \pm 2.7\%$ from Gemma 3. The low transfer from Gemma 3 is particularly noteworthy: despite Gemma 3's high self-ASR ($68.7\%$), attacks evolved against it do not generalize to Gemma 4, suggesting that Gemma 4's safety improvements address *fundamental* weaknesses rather than surface-level patches.

**Backward transfer is substantial and uniform.** Attacks from all generations transfer backward to Gemma 1 at ∼49–52% (Gemma 2: $49.0\%$; Gemma 3: $52.2\%$; Gemma 4: $48.9\%$). This indicates that newer evolutionary probes discover attacks exploiting longstanding vulnerabilities, not merely generation-specific weaknesses.

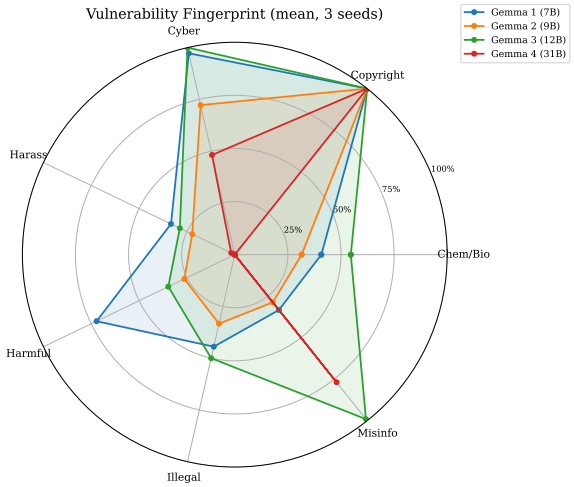



Figure 3: Vulnerability fingerprints across Gemma generations (mean ASR per category, 3 seeds). Gemma 2 and Gemma 4 show compact profiles; Gemma 3's profile is dramatically expanded in cybercrime and misinformation.

### 5.3 Robustness Across Generations

Figure 2 plots ASR and robustness ($1 - \text{ASR}$) against parameter count with error bars. We do not fit a scaling law to four data points. Instead, we observe that three of the four models (Gemma 1, 2, 4) are consistent with a gradual improvement trajectory, while Gemma 3 represents a clear outlier. This non-monotonic pattern coincides with the architectural transition from text-only (Gemma 1, 2) to multimodal (Gemma 3, 4), though Gemma 4's recovery shows that the regression is not an inherent consequence of multimodal design.

We document a non-monotonic robustness trajectory within one model family; we do *not* claim a general safety-vs-size scaling law.

### 5.4 Vulnerability Fingerprints

Figure 3 presents vulnerability fingerprints as radar charts. Each model exhibits a distinct profile:

- **Gemma 1**: moderate vulnerability with peaks in cybercrime ($97.2\%$) and copyright ($100\%$).
- **Gemma 2**: the most compact profile overall, though copyright remains at $100\%$ and cybercrime at $72.2\%$.
- **Gemma 3**: dramatically expanded in misinformation ($99.1\%$) and cybercrime ($100\%$). Only harassment shows continued improvement.
- **Gemma 4**: compact with the lowest overall ASR, but copyright persists at $100\%$.

## 6 Discussion

**Safety regression in Gemma 3.** The most surprising finding is Gemma 3's safety regression, which is statistically significant versus Gemma 2 ($p = 0.030$, paired bootstrap across 3 seeds). We hypothesize this is related to the introduction of multimodal capabilities: Gemma 3 was the first Gemma generation with vision-language support (architecture: `Gemma3ForConditionalGeneration`), requiring substantial architectural changes that may have disrupted safety-trained text representations. The fact that harassment, the category most explicitly targeted by safety training across all LLMs, continued to improve even in Gemma 3 supports this interpretation: category-specific safety training was maintained, but general robustness was compromised by the architectural shift.

**Gemma 4's recovery.** Gemma 4 achieved the highest robustness in the family ($33.9\% \pm 1.4\%$ ASR) despite being multimodal, suggesting that the safety regression in Gemma 3 was addressable. The

extremely low transfer rate from Gemma 3 to Gemma 4 ($14.0\% \pm 2.7\%$) indicates that Gemma 4's improvements were not simply patches for known Gemma 3 attacks but rather fundamental alignment improvements.

**Persistent category-level weaknesses.** Copyright stands out as a category that *no* Gemma generation has addressed: 100% ASR across all four models and all 9 seed-model combinations. This suggests a fundamental gap in safety training for intellectual property-related behaviors. Cybercrime similarly resists alignment in all but Gemma 4 ($33.3\%$).

**What we claim and what we do not.** We document a non-monotonic robustness trajectory within one model family; we do *not* claim a general safety-vs-size scaling law. With four data points, we cannot distinguish a U-shaped curve from noise. What we *can* say is that the Gemma 2 $\rightarrow$ Gemma 3 regression is statistically significant and large ($+23.2$ percentage points), and that the transfer matrix provides converging evidence (Gemma 3 is the easiest transfer target from all directions).

**Implications for safety evaluation.** Our findings have three implications:

1. **Static benchmarks are insufficient.** A single-timepoint benchmark would miss the non-monotonic safety trajectory entirely. Longitudinal evaluation with adaptive probes is necessary.
2. **Architectural changes require safety re-validation.** The Gemma 3 regression coincides with the multimodal architectural shift, suggesting that capability expansions should trigger comprehensive safety re-evaluation.
3. **Transfer analysis reveals alignment depth.** The distinction between Gemma 3 (high self-ASR, high inbound transfer) and Gemma 4 (low self-ASR, low inbound transfer) is only visible through cross-generational transfer analysis. A static benchmark applied to Gemma 3 alone would report high ASR but could not determine whether this reflects shallow or fundamental vulnerability.

**Limitations.**

- Three seeds per model provide reproducibility estimates but limited statistical power. The Gemma 3 $\rightarrow$ Gemma 4 improvement, while large ($-34.8$ pp), does not reach $p < 0.05$. Additional seeds would tighten confidence intervals.
- Single model family (Gemma) for generational analysis; findings may not generalize to other families (Claude, GPT, Llama).
- HarmBench coverage: 400 behaviors across 7 categories is comprehensive but not exhaustive.
- Evolutionary probe explores a subset of the attack space; absence of a successful attack does not prove safety.
- We study only the text modality; Gemma 3 and 4's vision capabilities may present additional attack surfaces not captured here.

# 7 Conclusion

We have presented a methodology for longitudinal safety evaluation of LLMs using quality-diversity evolution as a differential probe. Applied to four generations of Google's Gemma models with 3 independent seeds per model, we find that:

1. Safety alignment does not improve monotonically across model generations. Gemma 3 represents a statistically significant regression ($68.7\% \pm 4.7\%$ ASR vs. Gemma 2's $45.5\% \pm 5.9\%$; $p = 0.030$).
2. Cross-generational attack transfer is generation-dependent: attacks from any source transfer to Gemma 3 at $\sim$45% but only 14–18% to Gemma 4, suggesting Gemma 4's improvements are fundamental rather than superficial.

3. Vulnerability profiles are category-dependent: copyright remains at 100% ASR across all generations, while harassment shows consistent improvement and misinformation exhibits a 71× regression in Gemma 3 before recovering in Gemma 4.

These results underscore the need for *adaptive, longitudinal* safety evaluation. Static benchmarks provide a snapshot; evolutionary probing with transfer analysis reveals the *trajectory* of safety alignment. We advocate for model developers to publish generational safety analyses alongside capability benchmarks, and for the research community to develop standardized protocols for cross-version safety comparison.

# References


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# NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]The abstract states two specific research questions (cross-generational transfer and safety scaling) and the introduction enumerates four concrete contributions, each tied to a results section.
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]Section 6 includes a dedicated Limitations paragraph covering: small N for scaling, single model family, HarmBench coverage bounds, LLM judge bias, and evolutionary probe coverage limitations.
3. **Theory Assumptions and Proofs**
   (a) For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof? [N/A]This is an empirical paper with no theoretical results.
4. **Experimental Result Reproducibility**
   (a) Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)? [Yes]Section 4 specifies all models, hyperparameters (grid dimensions, batch size, generations, concurrency), seed dataset (HarmBench 400 behaviors), judge model (GPT-5.4-mini), serving infrastructure (vLLM 0.19.1, bf16, temperature 0), and hardware (H200, RTX 4090). The transfer replay protocol is fully described in Section 3.2.
5. **Open access to data and code**
   (a) Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material? [Yes]Anonymous code repository is provided as supplemental material, including the full evolution framework (Rust), analysis scripts (Python), orchestration scripts, and instructions for reproducing all experiments. Seed behaviors are from the public HarmBench dataset. Experiment result JSONs are included.
6. **Experimental Setting/Details**
   (a) Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results? [Yes]All MAP-Elites hyperparameters are specified in Section 4. No training is performed; we evaluate existing models. Evolution configuration: 200 generations, batch size 50, $6 \times 6 \times 8$ MAP-Elites grid, mutation rate 0.3, crossover rate 0.7.
7. **Experiment Statistical Significance**
   (a) Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments? [Yes]All experiments use 3 independent random seeds per model. We report mean $\pm$ standard deviation and 95% bootstrap confidence intervals. Paired bootstrap tests are used for pairwise model comparisons ($p < 0.05$ threshold). We acknowledge that $N = 3$ provides limited power for some contrasts.
8. **Experiments Compute Resources**
   (a) For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time for each experiment, etc.) needed to reproduce the experiments? [Yes]Section 4 reports: NVIDIA H200 NVL (141 GB), vLLM 0.19.1, total GPU time $\sim$55 hours across two machines. Frontier models via proprietary APIs.
9. **Code Of Ethics**
   (a) Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics? [Yes]This work studies LLM safety alignment to improve defenses.

We follow responsible disclosure practices: we do not release specific jailbreak prompts that could enable misuse, only aggregate statistics and methodology. The goal is to help model developers identify and fix vulnerabilities.

10. **Broader Impacts**

    (a) Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed? [Yes]Positive: enables systematic safety evaluation across model generations, helping developers prioritize alignment work. Negative: methodology could in principle be used to find and exploit vulnerabilities. We mitigate this by not publishing specific attack prompts and by framing the work as a defensive evaluation tool.

11. **Safeguards**

    (a) Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)? [Yes]We release aggregate statistics (transfer rates, ASR per category) and the evolution framework, but not the specific attack prompts or model responses from our experiments. The seed behaviors come from the publicly available HarmBench dataset.

12. **Licenses for existing assets**

    (a) Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected? [Yes]HarmBench is cited and used under its original license. Gemma models are used under Google's Gemma license. Proprietary models are accessed via their official APIs under standard terms of service.

13. **New Assets**

    (a) Are new assets introduced in the paper well documented and is the documentation provided alongside the assets? [Yes]The evolution framework is documented with README, CLI help text, and reproduction instructions. Analysis scripts include docstrings and usage examples.

14. **Crowdsourcing and Research with Human Subjects**

    (a) For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation? [N/A]No human subjects or crowdsourcing.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    (a) Does the paper describe potential risks incurred by study participants, whether combative or non-combative to the subjects, and does it meet the necessary IRB criteria? [N/A]No human subjects.

16. **Declaration of LLM usage**

    (a) Does the paper describe the usage of LLMs in the research and/or writing process? [Yes]LLMs are used in two capacities: (1) as evaluation targets (the Gemma models being probed for safety), and (2) as a self-hosted automated judge (Llama-3.1-8B-Instruct scoring target responses). An LLM coding assistant was used to help with framework development and analysis scripts.